

What the corpus costs

Per-episode generation cost measured on real, EdenAI-billed episodes across three teachers and four multi-hop QA datasets – with projections at 1K, 10K and 120K episodes, and a costed plan for a 12,000-episode mixed corpus.

Date 2026-08-30 Billing EdenAI, per call EUR→USD 1.1607 Episodes measured 26
Cost of this study ≈ \$2.40

\$0.0425–0.0947 COST PER EPISODE
€665 12,000-EPISODE MIX
€865 SAME, WITH CONTINGENCY
€0 SAME CORPUS ON FAU

01 Basis of estimate

Every figure comes from episodes actually run and actually billed – EdenAI returns a real USD `cost` per call, which is summed across every logged LLM call in an episode. Nothing here is a list-price extrapolation.

Configuration	g3_plan — guidance level 3, plan review (3 rounds), step budget 3, budget hidden
Student	fau/ibm-granite/granite-4.1-3b — free FAU gateway, contributes \$0
Teacher endpoint	https://api.edenai.run/v3/chat/completions
Sample	5 episodes per teacher (HotpotQA) · 3 episodes per dataset (Qwen3.8)

A teacher-guidance episode costs **\$0.042–\$0.095**. Per-episode spend is set by the protocol – roughly four teacher calls per episode – far more than by question difficulty. The teacher is effectively 100% of the bill; the student is free.

02 Measured cost per episode

By teacher – HotpotQA, n=5 each

TEACHER	EDENAI MODEL ID	\$/EPISODE	SD	CALLS	API TIME	CORRECT
Qwen3.8-2.4T-A95B	qwen/qwen3.8-2.4t-a95b	\$0.0448	±0.0134	4.2	200 s	4/5
Kimi K3	qwen/kimi-k3	\$0.0599	±0.0142	4.2	141 s	3/5
GLM-5.3 	deepinfra/zai-org/GLM-5.3	\$0.0800	±0.0232	9.2	407 s	4/5

 **GLM-5.3's figure reflects a misconfiguration, not its price.** At a 2,500-token cap it returns empty content with `finish_reason: length` – inline reasoning consumes the budget before the JSON verdict – so every call is retried (9.2 calls vs 4.2). An 8,000-token cap returns valid JSON for \$0.0137, versus \$0.0123 paid for nothing, but then exceeds the 180 s EdenAI timeout. Treat GLM-5.3 as an **upper bound** pending a corrected run.

By dataset – teacher Qwen3.8-2.4T-A95B

DATASET	N	\$/EPISODE	SD	HOPS	GOLD	LICENCE	CORRECT	GROUNDED
StrategyQA	3	\$0.0425	±0.0107	2–5	paragraph	MIT	2/3	1/3
HotpotQA	5	\$0.0448	±0.0134	2	sentence	CC BY-SA 4.0	4/5	5/5
MuSiQue	3	\$0.0470	±0.0097	2–4	paragraph	CC BY 4.0	1/3	3/3
2WikiMultihopQA	3	\$0.0530	±0.0038	2–4	sentence	Apache-2.0	1/3	1/3
Spread across datasets is only ±12%. 2Wiki is dearest because the teacher writes longer critiques on 4-hop comparisons (11.3k output tokens vs 9.3k).								

03 Cost per episode – teacher × dataset

Nine of the twelve cells are derived: a dataset's measured cost scaled by that teacher's HotpotQA ratio (Qwen 1.00, Kimi 1.34, GLM 1.79). Directly measured cells are marked .

\$/EPISODE	HOTPOTQA	2WIKIMULTIHOPQA	MUSIQUE	STRATEGYQA
Qwen3.8-2.4T-A95B	● \$0.0448	● \$0.0530	● \$0.0470	● \$0.0425
Kimi K3	● \$0.0599	\$0.0709	\$0.0629	\$0.0568
GLM-5.3	● \$0.0800	\$0.0947	\$0.0840	\$0.0759

04

Scale scenarios

Single teacher, dataset-balanced. **Plan** adds a 30% contingency over the measured mean, matching the largest observed per-episode standard deviation. Budget against the Plan figure.

TEACHER	1,000	10,000	120,000
Qwen3.8-2.4T-A95B	\$46.79 · plan \$60.83	\$467.93 · plan \$608.31	\$5,615 · plan \$7,300
Kimi K3	\$62.60 · plan \$81.38	\$625.98 · plan \$813.77	\$7,512 · plan \$9,765
GLM-5.3	\$83.62 · plan \$108.71	\$836.24 · plan \$1,087	\$10,035 · plan \$13,045

In EUR, on the cheapest teacher: **1K ≈ €40** (plan €52) · **10K ≈ €403** (plan €524) · **120K ≈ €4,838** (plan €6,289).

05

The mixed corpus — 1,000 per teacher × dataset

Three teachers × four datasets × 1,000 episodes = **12,000 episodes**, the design that makes teacher identity and dataset both analysable as variables.

USD	HOTPOTQA	2WIKI	MUSIQUE	STRATEGYQA	ROW TOTAL
Qwen3.8-2.4T-A95B	\$44.75	\$52.98	\$46.99	\$42.45	\$187.17
Kimi K3	\$59.87	\$70.87	\$62.86	\$56.79	\$250.39
GLM-5.3	\$79.98	\$94.68	\$83.97	\$75.87	\$334.50
Column total	\$184.60	\$218.53	\$193.82	\$175.11	\$772.06

12,000 EPISODES	USD	EUR
Measured mean	\$772.06	€665.16
Planning figure (+30%)	\$1,003.68	€864.70

Budget €865 for the 12,000-episode mixed corpus; expect $\approx$ €665. Fixing GLM-5.3's token cap should cut its row materially, taking the total toward €550–600.

06 Cost levers

- **Run bulk generation on the free FAU gateway.** `DeepSeek-V4-Flash` and `gpt-oss-120b` were both verified working and free on 2026-08-30. Paid teachers are only needed where teacher identity is the variable under study. This is the difference between €865 and near-zero for the bulk corpus.
- **Fix GLM-5.3 before buying any of it.** It needs an 8,000-token cap and an EdenAI timeout $\geq$ 600 s. It expands its reasoning to fill whatever cap it is given (16k $\rightarrow$ 5,943 tokens, \$0.0243), so 8,000 is a sweet spot, not a floor.
- **Step budget drives cost roughly linearly.** These figures are budget 3; plan a budget-5 run on the deeper datasets at $\sim$ \$0.07–0.08 per episode.
- **Wall-clock may bind before money does.** At $\sim$ 200 s of teacher API time per episode, 12,000 episodes is $\sim$ 660 hours serialized. EdenAI concurrency limits are unmeasured.

07 Caveats

- **n is small** – five episodes per teacher, three per dataset. The 30% contingency exists for this reason.
- **Nine of twelve matrix cells are derived**, not measured.
- **GLM-5.3 is an upper bound**, pending a corrected run.
- **Student deviation:** granite-4.1-3b on FAU, not local granite-4.2 – every GPU node was draining for a reboot, and 4.2 is not in the FAU catalogue. Student calls are free either way, so this affects trajectory shape, not cost.
- **Accuracy is not a cost figure** but is reported because it decides how many episodes are needed: 4/5 correct on HotpotQA against 1/3 on 2Wiki and MuSiQue.

08 Resources

Endpoints

EdenAI · docs.edenai.co

`api.edenai.run/v3/chat/completions`

NHR@FAU gateway (free) · hpc.fau.de

`hub.nhr.fau.de/api/llmgw/v1`

Models

Qwen · `qwen/qwen3.8-2.4t-a95b`

moonshotai · `qwen/kimi-k3`

zai-org · `deepinfra/zai-org/GLM-5.3`

ibm-granite/granite-4.1-3b — student

Datasets

HotpotQA — CC BY-SA 4.0

2WikiMultihopQA — Apache-2.0

MuSiQue — CC BY 4.0

StrategyQA — MIT · dataset zip

Data behind this report

`reports/costprobe_teachers.json`

`reports/costprobe_datasets.json`

`reports/budget/budget_data.json`

`scripts/episode_cost_report.py`

`scripts/build_budget_report.py`

Exchange rate

open.er-api.com — 1.1607, 2026-08-30

Teacher-Guidance corpus budget · measured 2026-08-30 · all costs are EdenAI per-call billing on real episodes

Repository: `/shared/al-maamari/DeKIS/teacher-guidance` · branch `feature/teacher-guidance`